The Wayback Machine - https://web.archive.org/web/20250423172254/https://iowow.softmotions.com/

C11 key/value database engine

https://github.com/Softmotions/iowow (https://web.archive.org/web/20250423172254/https://github.com/Softmotions/iowow)

IOWOW - The C11 persistent key/value storage based on **skip list (https://web.archive.org/web/20250423172254/https://en.wikipedia.org/wiki/Skip_list)** data structure

Features

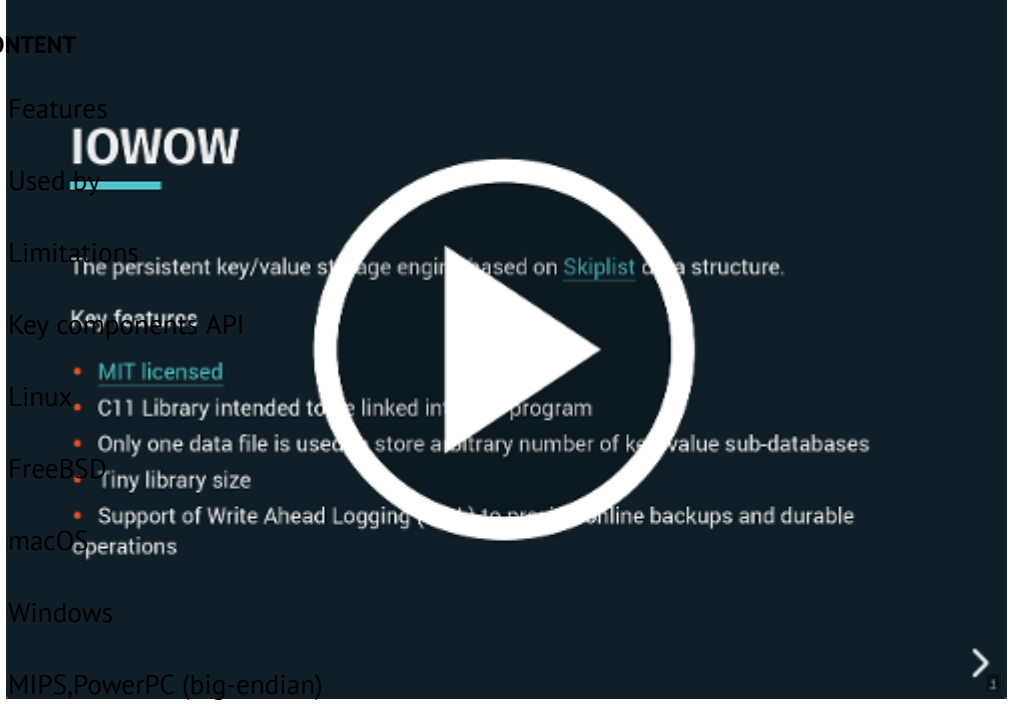
- Support of multiple key-value databases within a **single file**
- Online database backups
- Native support of integer keys
- Write Ahead Logging (WAL) (/web/20250423172254/https://iowow.softmotions.com/wal)
- Ultra-fast traversal of database records
- Compound keys support
- Good performance comparing its main competitors: lmdb (https://web.archive.org/web/20250423172254/https://symas.com/lmdb/), leveldb (https://web.archive.org/web/20250423172254/https://github.com/google/leveldb), kyoto cabinet (https://web.archive.org/web/20250423172254/http://fallabs.com/kyotocabinet/)
- Tiny C11 library (only 200K) can be easily embedded into any software

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(/web/20250423172254/https://iowow.softmotions.com/rs/mw/link/Https://iowow.softmotions.com/articles/intro/)

Used by

- EJDB - Embeddable JSON database engine. http://ejdb.org (https://web.archive.org/web/20250423172254/http://ejdb.org/)

Limitations

- Maximum storage file size: `512 GB (0x7fffffff80)` (since v1.0.6)
- The total size of a single key+value record must not be greater than `255Mb (0xfffffff)`

Key components API

- iwkv.h (https://web.archive.org/web/20250423172254/https://github.com/Softmotions/iowow/blob/master/src/kv/iwkv.h) Persistent key/value database engine: `iwkv`
- iwfsmfile.h (https://web.archive.org/web/20250423172254/https://github.com/Softmotions/iowow/blob/master/src/fs/iwfsmfile.h) File blocks allocation manager like `malloc()` on files

Supported platforms

Linux

Building debian packages

CONTENT

```
umask 022
mkdir build && cd build
cmake .. -DCMAKE_BUILD_TYPE=Release -DPACKAGE_DEB=ON
make package
```

RPM based Linux distributions

```
umask 022
mkdir build && cd build
cmake .. -DCMAKE_BUILD_TYPE=Release -DPACKAGE_RPM=ON
make package
```

FreeBSD

Successfully tested on FreeBSD 10/11

https://www.freshports.org/databases/iowow/ (https://web.archive.org/web/20250423172254/https://www.freshports.org/databases/iowow/)

macOS

Successfully tested on OSX 10.12/10.13

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Windows

Cross-compilation for Windows x86-64 platform (/web/20250423172254/https://iowow.softmotions.com/iw/win)

MIPS,PowerPC (big-endian)

Successfully tested on

- Debian 9.4, MIPS 32, gcc 6.x compiler.
- PowerPC FreeBSD

Benchmarks

https://iowow.softmotions.com/articles/intro/#/_benchmarks (https://web.archive.org/web/20250423172254/https://iowow.softmotions.com/articles/intro/#/_benchmarks)

https://iowow.softmotions.com/articles/intro/benchmark_results.html (https://web.archive.org/web/20250423172254/https://iowow.softmotions.com/articles/intro/benchmark_results.html)

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```c
#include <iowow/iwkv.h>
#include <string.h>
#include <stdlib.h>

int main() {
  IWKV_OPTS opts = {
    .path = "example1.db",
    .oflags = IWKV_TRUNC // Cleanup database before open
  };
  IWKV iwkv;
  IWDB mydb;
  iwrc rc = iwkv_open(&opts, &iwkv);
  if (rc) {
    iwlog_ecode_error3(rc);
    return 1;
  }
  // Now open mydb
  // - Database id: 1
  rc = iwkv_db(iwkv, 1, 0, &mydb);
  if (rc) {
    iwlog_ecode_error2(rc, "Failed to open mydb");
    return 1;
  }
  // Work with db: put/get value
  IWKV_val key, val;
  key.data = "foo";
  key.size = strlen(key.data);
  val.data = "bar";
  val.size = strlen(val.data);

  fprintf(stdout, "put: %.*s => %.*s\n",
          (int) key.size, (char *) key.data,
          (int) val.size, (char *) val.data);

  rc = iwkv_put(mydb, &key, &val, 0);
  if (rc) {
    iwlog_ecode_error3(rc);
    return rc;
  }
  // Retrive value associated with `foo` key
  val.data = 0;
  val.size = 0;
  rc = iwkv_get(mydb, &key, &val);
  if (rc) {
    iwlog_ecode_error3(rc);
    return rc;
  }

  fprintf(stdout, "get: %.*s => %.*s\n",
          (int) key.size, (char *) key.data,
          (int) val.size, (char *) val.data);

  iwkv_val_dispose(&val);
  iwkv_close(&iwkv);
  return 0;
}
```

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Compile and run

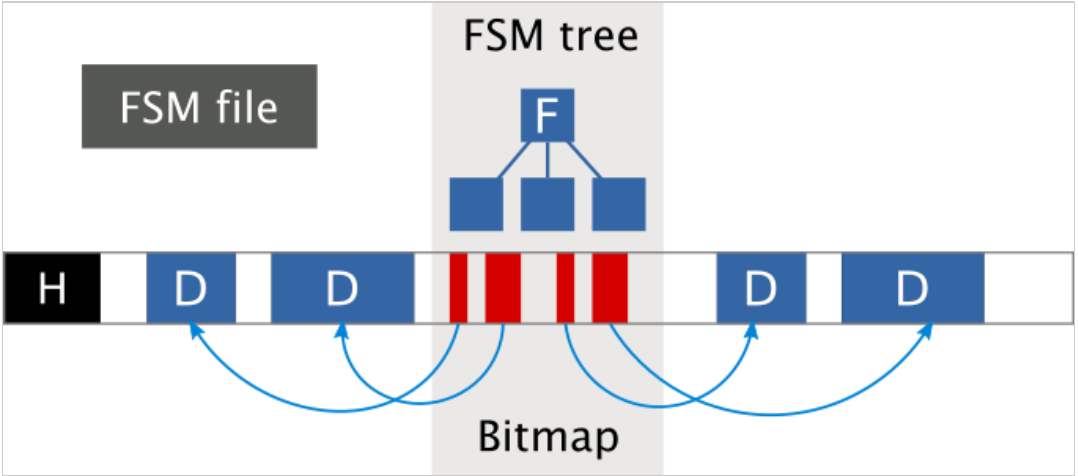
```
gcc -std=gnu11 -Wall -pedantic -c -o example1.o example1.c
gcc -o example1 example1.o -liowow

./example1
put: foo => bar
get: foo => bar
```

Internals

High level overview

- IWKV databases stored in a single file. Free space managed by bitmap (https://web.archive.org/web/20250423172254/https://en.wikipedia.org/wiki/Free_space_bitmap) index (iwfsmfile.h (https://web.archive.org/web/20250423172254/https://github.com/Softmotions/iowow/blob/master/src/fs/iwfsmfile.h)).
- All free space chunks stored in memory BTree index. BTree index constructed from bitmap after iwkv file opened.
- Minimal allocation amount: `128 bytes`
- Max number of skip list levels: `24`
- Write Ahead Log (WAL) (/web/20250423172254/https://iowow.softmotions.com/wal)



(/web/20250423172254/https://iowow.softmotions.com/rs/mw/link/Image:/2202/fsm.png)
Free space map file (FSM)

- `H` FSM file header custom user header together with auxiliary system data.
- `D` Data block. Data block is allocated chunk of file address space whose allocation state stored in free space bitmap (https://web.archive.org/web/20250423172254/https://en.wikipedia.org/wiki/Free_space_bitmap)
- `F` Free space map tree

IWKV storage pool consists mostly of two kinds of blocks:

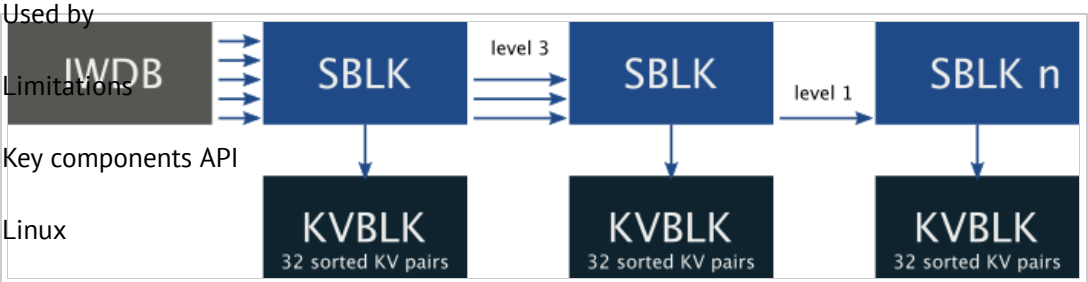
- `SBLK` Skip list node with pointers to next nodes and pointer to `KVBLK` (key/value pairs block). SBLK has fixed size (256 bytes). SBLK file position

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(block address) within a file is fixed and cannot be changed.

- `KVBLK` Storage block for a set of key value pairs. Up to 32 adjacent KV pairs can be stored within a single KVBLK block.



(/web/20250423172254/https://iowow.softmotions.com/rs/mw/link/Image:/2203/iwkv1.png)

IWKV skip list + key value blocks

SBLK

MIPS,PowerPC (big-endian)

```
[flags:u1,lvl:u1,lkl:u1,pnum:u1,p0:u4,kblk:u4,pi:u1[32],n:u4[24],lk:u1
                                                  \
                                                 KVBLK
```

1. `flags` Persistent block flags (1 byte)
2. `lvl` Skiplist level of this block (1 byte)
3. `lkl` Length of the lowest key prefix in this block (1 byte)
4. `kblk` Block number of associated `KVBLK` (4 bytes)
5. `pi[32]` Array of key/value pair indexes in `KVBLK` block. Indexes are sorted by keys. (32 bytes)
6. `n[24]` Pointers to next `SBLK` blocks in skiplist (96 bytes)
7. `lk` Buffer for the lowest key prefix among all key/value pairs stored in `KVBLK`

KVBLK

```
[szpow:u1,idxsz:u2,KVI[32] ___free space___ [[KV],...]]
```

1. `szpow` KVBLK length as power of 2
2. `idxsz` Length of KVI array in bytes
3. `KVI[32]` Key/value pair index `[ps:vn,pl:vn]`
   `ps` Key/value pair block offset on *i-th* place variable length encoded number. This offset is relative to end of KVBLK block
   `pl` Key/value pair block length on *i-th* place variable length encoded number
4. `KV` Key/value pair `[klen:vn,key,value]`
   `klen` Key length as variable length encoded number
   `key` Key data buffer
   `value` Value data buffer

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